

DLD HW3 Spring 2026 Solution

Q.1)

In an octal system, the digits range from 0 to 7. The 7's complement of a digit n is calculated as $7 - n$. Since we only have 8 possible values (0–7), we only need 3 input bits (A, B, C) and 3 output bits (X, Y, Z) ($2^3 = 8$).

Decimal	A (MSB)	B	C (LSB)	X	Y	Z
0	0	0	0	1	1	1
1	0	0	1	1	1	0
2	0	1	0	1	0	1
3	0	1	1	1	0	0
4	1	0	0	0	0	1
5	1	0	1	0	0	1
6	1	1	0	0	0	0
7	1	1	1	0	0	0

A \ B C	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
$\bar{A}$	0	1	3	2
A	4	5	7	6

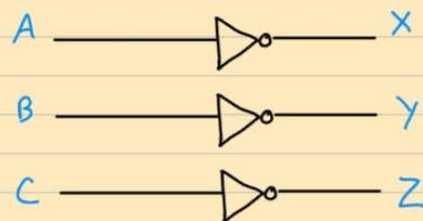
$$X = \bar{A}$$

A \ B C	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
$\bar{A}$	0	1	3	2
A	4	5	7	6

$$Y = \bar{B}$$

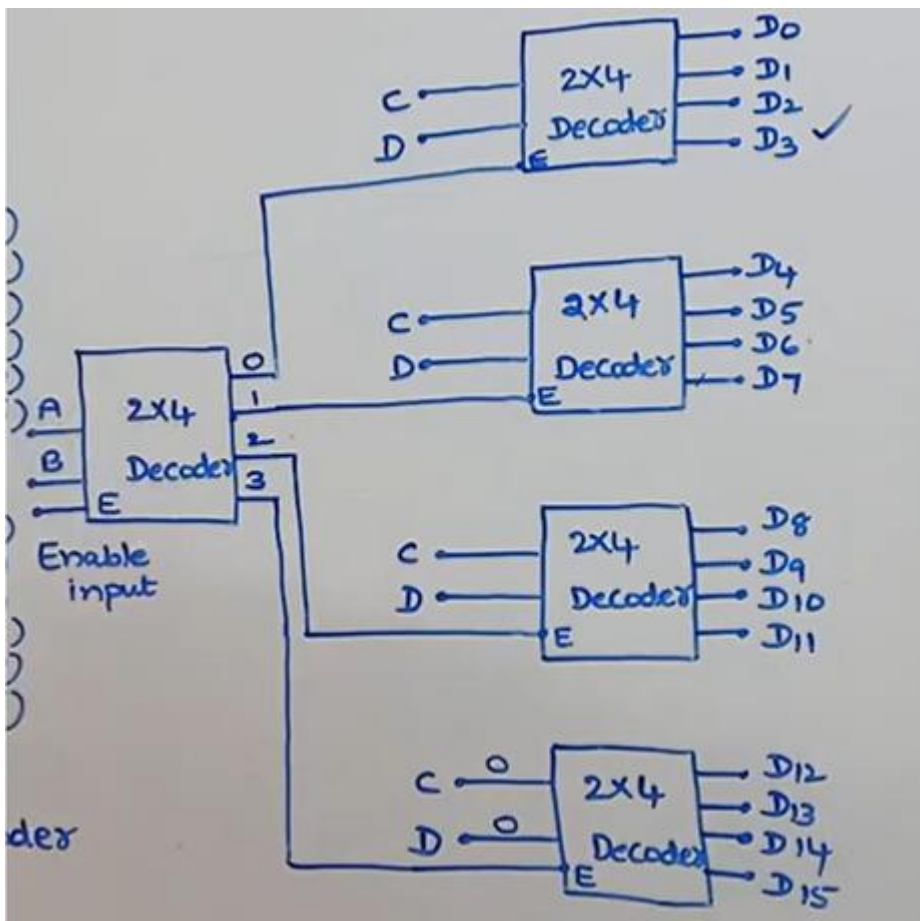
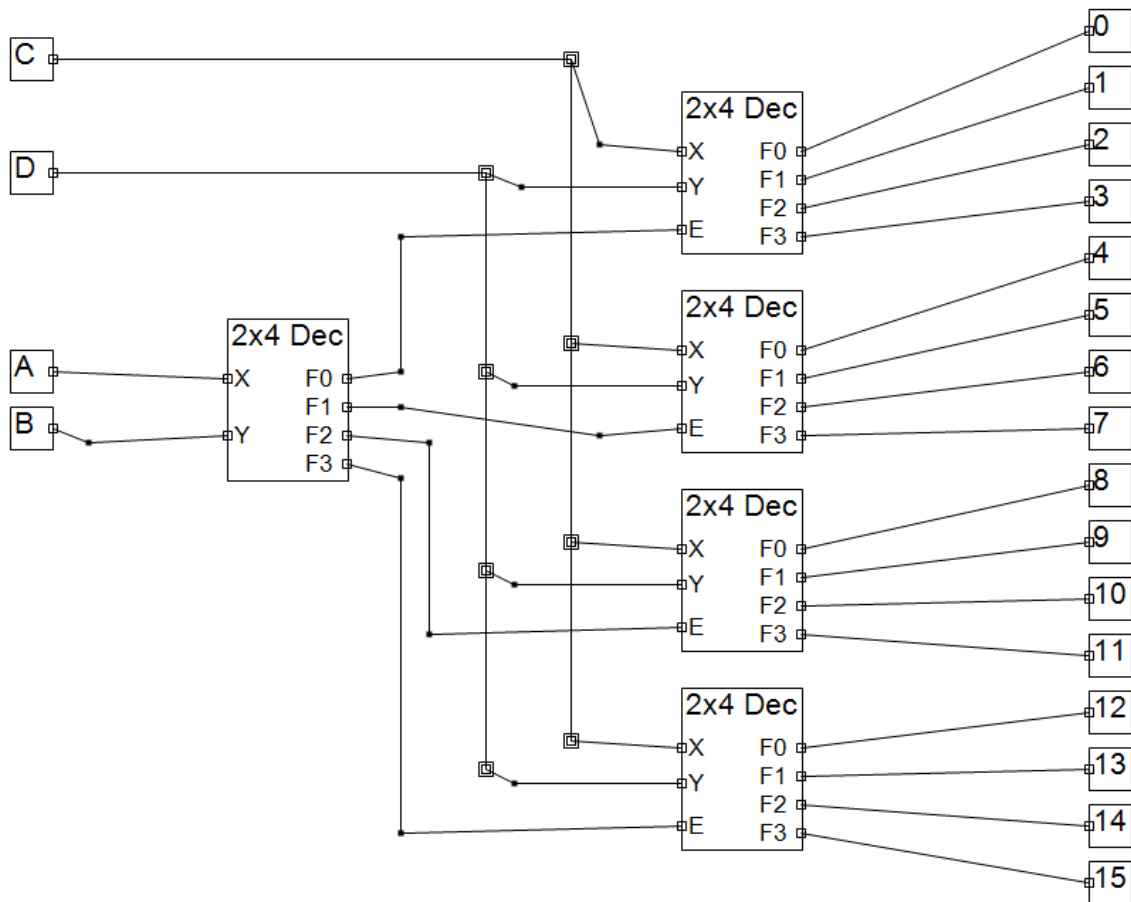
A \ B C	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
$\bar{A}$	0	1	3	2
A	4	5	7	6

$$Z = \bar{C}$$



Q.2)

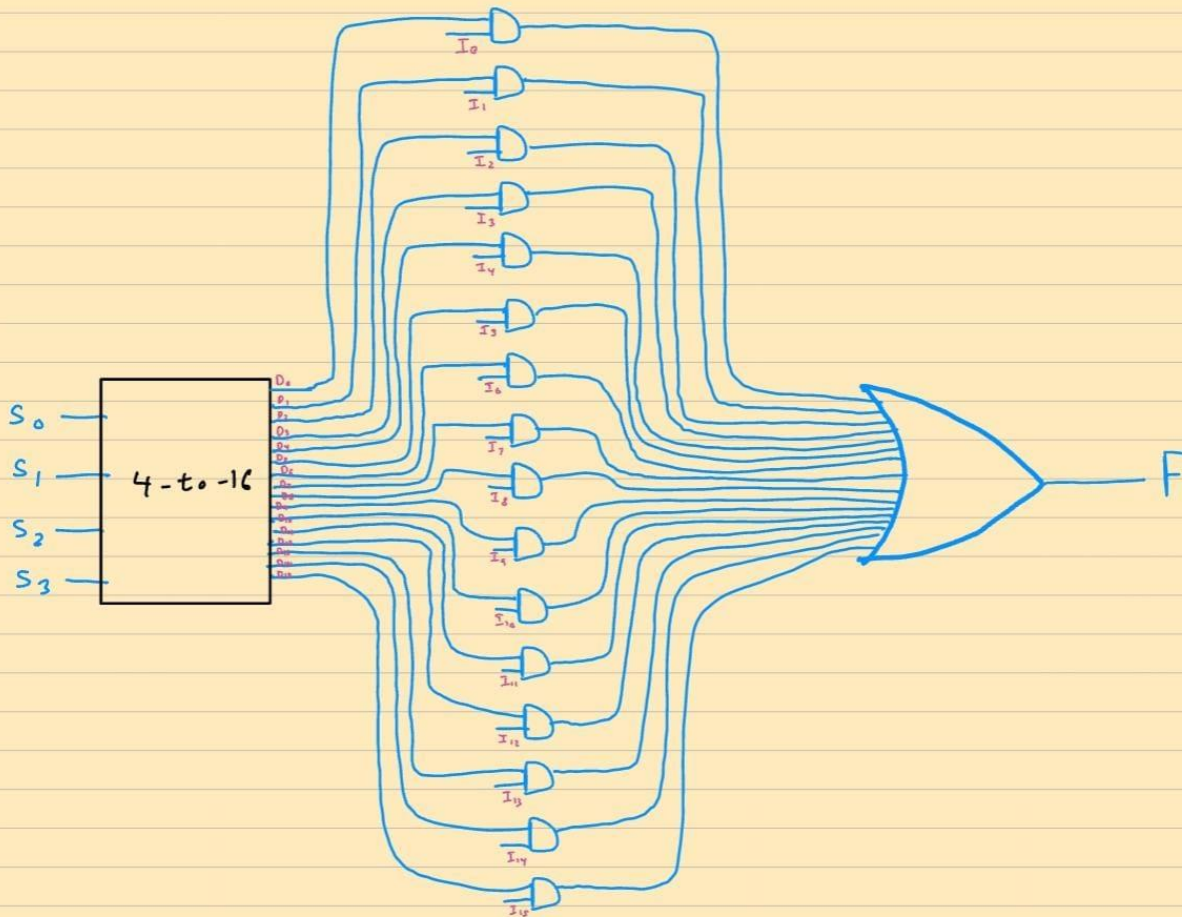
a)



b)

16-to-1 MUX with decoder + logic gates

Using a 4-to-16 decoder since our mux will have 16 inputs



Whatever signal is present on $I_0 \rightarrow I_{15}$ will be transmitted to the output OR gate depending on the states of S_0, S_1, S_2, S_3 which will act as second input to and gates.

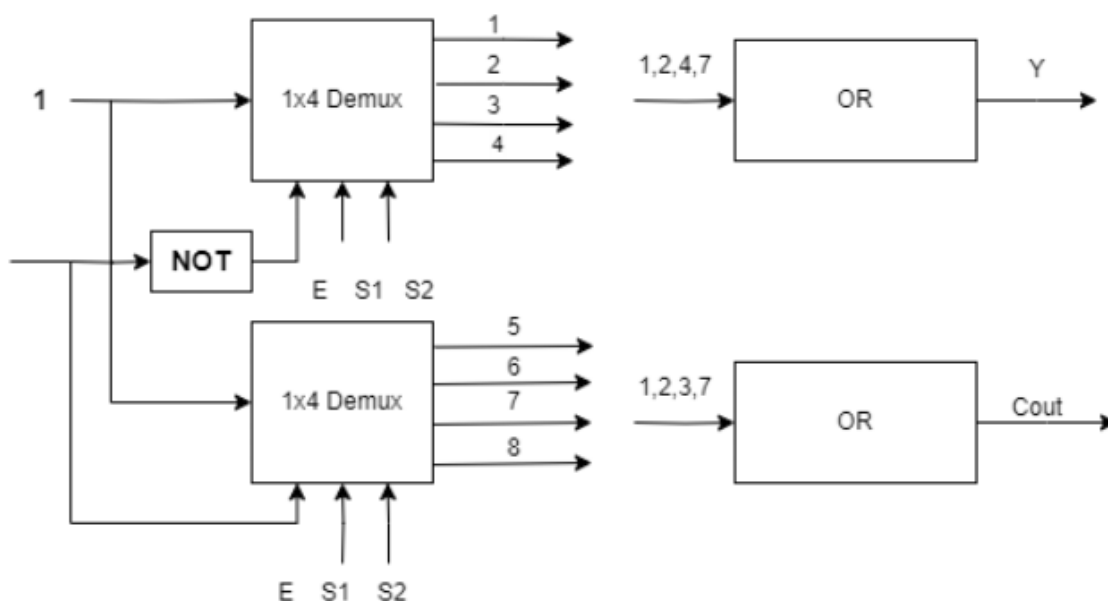
c)

A	B	Cin	Y	Cout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$Y = \sum (1,2,4,7)$$

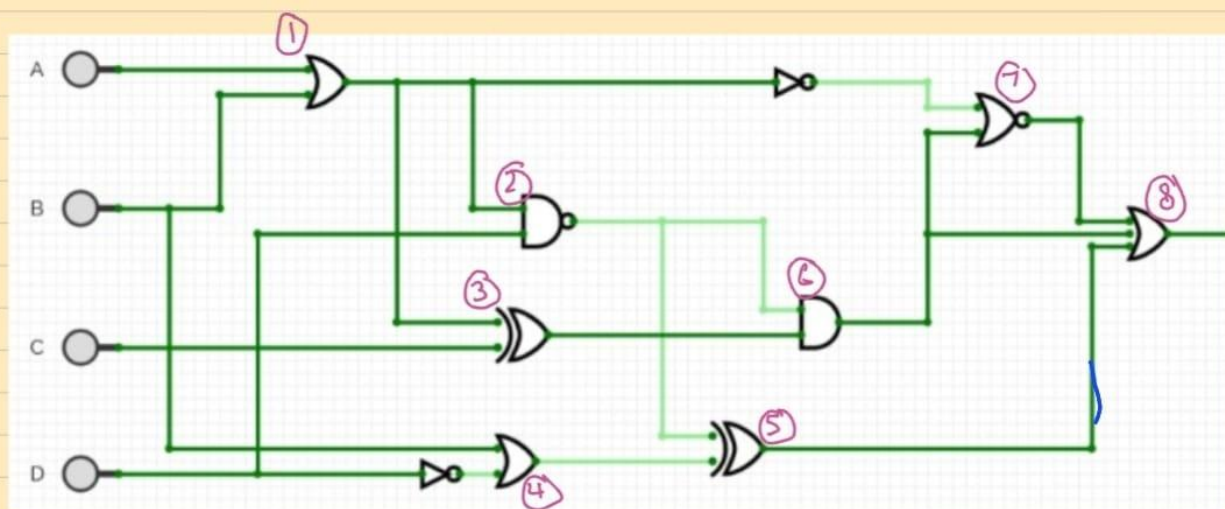
$$Cout = \sum (1,2,3,7)$$

A	B	Cin	F0	F1	F2	F3	F4	F5	F6	F7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1



Q.3)

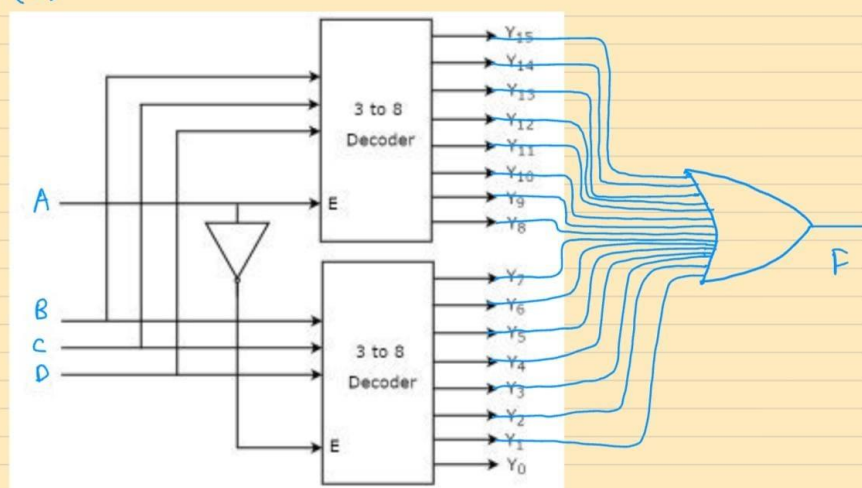
a)



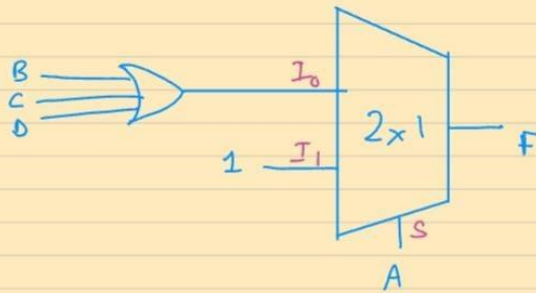
$$F(A, B, C, D) = \sum m(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)$$

A	B	C	D	Gate-1	Gate-2	Gate-3	D'	Gate-4	Gate-5	Gate-6	(Gate-1)'	Gate-7	Gate-8
0	0	0	0	0	1	0	1	1	0	0	1	0	0
0	0	0	1	0	1	0	0	0	1	0	1	0	1
0	0	1	0	0	1	1	1	1	0	1	1	0	1
0	0	1	1	0	1	1	0	0	1	1	1	0	1
0	1	0	0	1	1	1	1	1	0	1	0	0	1
0	1	0	1	1	0	1	0	1	1	0	0	1	1
0	1	1	0	1	1	0	1	1	0	0	0	1	1
0	1	1	1	1	0	0	0	1	1	0	0	1	1
1	0	0	0	1	1	1	1	1	0	1	0	0	1
1	0	0	1	1	0	1	0	0	0	0	0	1	1
1	0	1	0	1	1	0	1	1	0	0	0	1	1
1	0	1	1	1	0	0	0	0	0	0	0	1	1
1	1	0	0	1	1	1	1	1	0	1	0	0	1
1	1	0	1	1	0	1	0	1	1	0	0	1	1
1	1	1	0	1	1	0	1	1	0	0	0	1	1
1	1	1	1	1	0	0	0	1	1	0	0	1	1

(b)

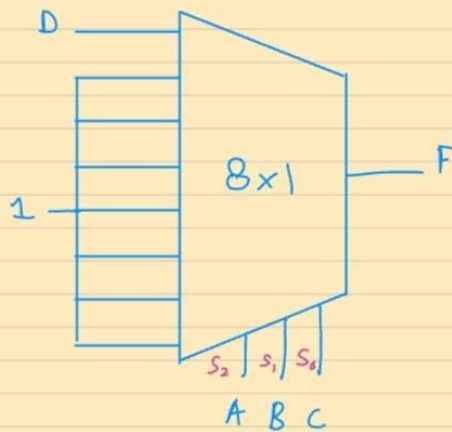


(c)



Output is 1 except when B, C, D are all 0

(d)



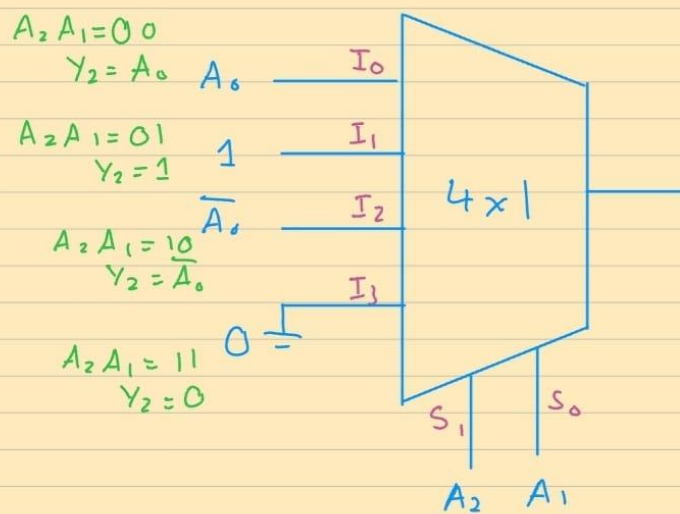
If $D=1$, $F=1$. If $D=0$, $F=0$

Output is High (1) for all other inputs.

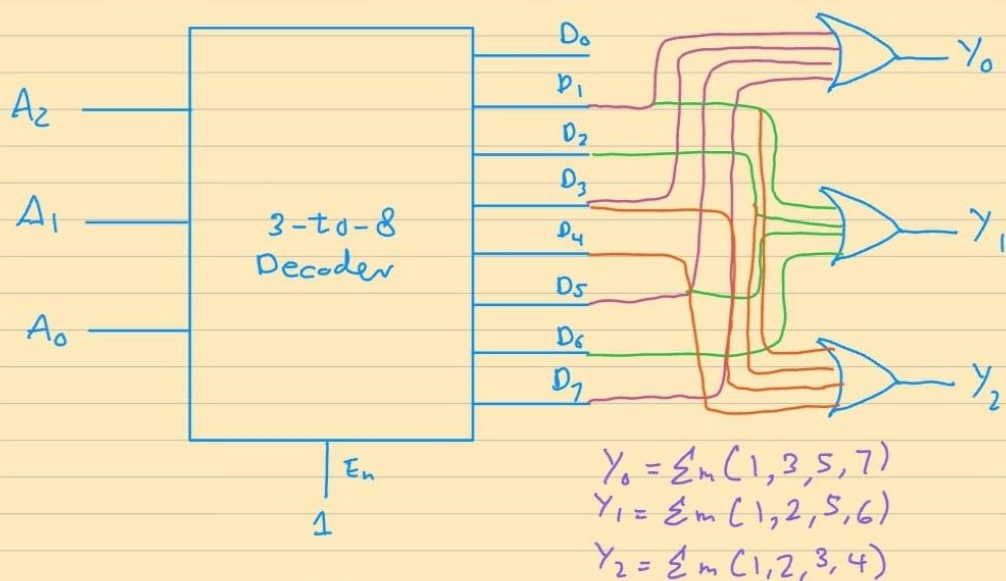
Q.4)

Decimal	A2	A1	A0	Y2	Y1	Y0
0	0	0	0	0	0	0
1	0	0	1	1	1	1
2	0	1	0	1	1	0
3	0	1	1	1	0	1
4	1	0	0	1	0	0
5	1	0	1	0	1	1
6	1	1	0	0	1	0
7	1	1	1	0	0	1

Q-4) (a)



(b)



MUX implementation :: it uses less gates and is more robust

Q.5)

a)

Weight Delta (mg)	A	B	C	X (Motor Brakes)	Y (Nozzle)	Z (Filler)
10	0	0	0	1	1	0
7.5	0	0	1	1	1	0
5	0	1	0	0	0	0
2.5	0	1	1	0	0	0
-2.5	1	0	0	0	0	0
0	1	0	1	0	0	0
-5	1	1	0	0	0	0
-10	1	1	1	1	0	1

b)

Q-5) (b)

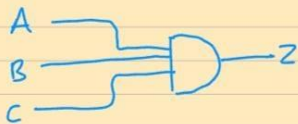
A \ B ^c	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
$\bar{A}$	0 1	1 1	3	2
A	4	5	7	6

$$Y = \bar{A}\bar{B}$$



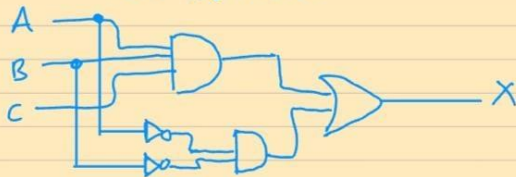
A \ B ^c	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
$\bar{A}$	0	1	3	2
A	4	5	7 1	6

$$Z = ABC$$



A \ B ^c	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
$\bar{A}$	0 1	1 1	3	2
A	4	5	7 1	6

$$X = \bar{A}\bar{B} + ABC$$



c)

Weight Delta (mg)	A (Ground)	B	C	X (Brakes)	Y (Nozzle)	Z (Filler)
10	0	0	0	1	1	0
7.5	0	0	1	1	1	0
5	0	1	0	0	0	0
2.5	0	1	1	0	0	0
-2.5 (Mapped to +10)	0	0	0	1	1	0
0 (Mapped to +7.5)	0	0	1	1	1	0
-5 (Mapped to +5)	0	1	0	0	0	0
-10 (Mapped to +2.5)	0	1	1	0	0	0

(c) $Y = \bar{B}$ (by observation)

$Z = 0$

$X = \bar{B}$

→ safety hazard: system does not activate the filler (Z)

→ underweight tablets will pass through the line w/o motor stopping or filler adding material

↳ violates req of neg deltas staying within 8 mg

→ false corrections: system may drill tablets that are 0 or -2.5 mg → by mistakenly identifying as overweight

Q.6)

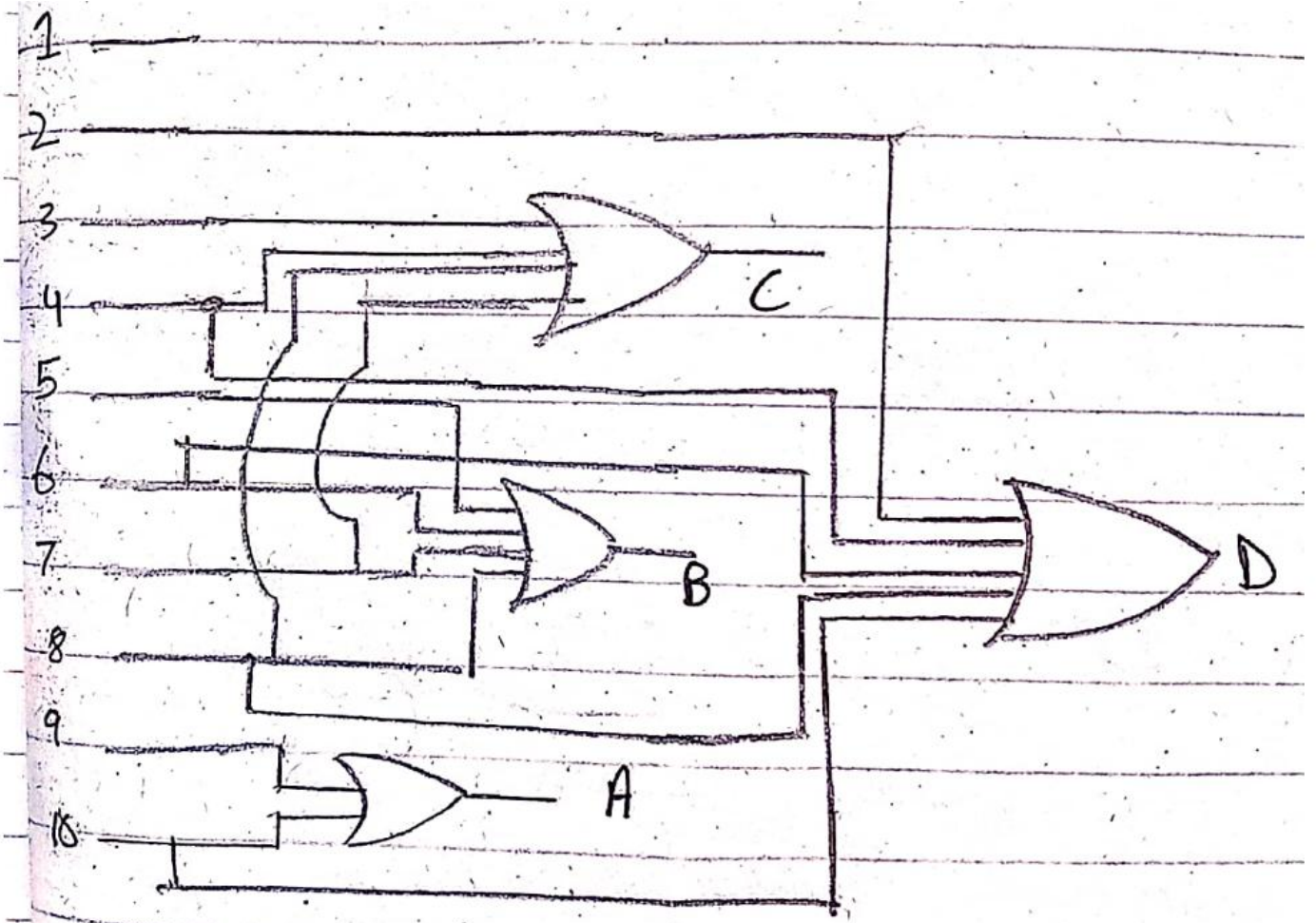
a)

I1	I2	I3	I4	I5	I6	I7	I8	I9	I10	A	B	C	D
1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	0	0	0	1	0	0
0	0	0	0	0	1	0	0	0	0	0	1	0	1
0	0	0	0	0	0	1	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	1	1	1
0	0	0	0	0	0	0	0	1	0	1	0	0	0
0	0	0	0	0	0	0	0	0	1	1	0	0	1

$$A = I_9 + I_{10} \quad , \quad B = I_5 + I_6 + I_7 + I_8$$

$$C = I_3 + I_4 + I_7 + I_8$$

$$D = I_2 + I_4 + I_6 + I_8 + I_{10}$$



b)

If more than 1 input is high output is ambiguous. Hence using priority encoder.

I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	I_{10}	A	B	C	D
0	0	0	0	0	0	0	0	0	X	0	0	0	0
1	0	0	0	0	0	0	0	0	X	0	0	0	1
X	1	0	0	0	0	0	0	0	X	0	0	1	0
X	X	1	0	0	0	0	0	0	X	0	0	1	1
X	X	X	1	0	0	0	0	0	X	0	1	0	0
X	X	X	X	1	0	0	0	0	X	0	1	0	1
X	X	X	X	X	1	0	0	0	X	0	1	1	0
X	X	X	X	X	X	1	0	0	X	0	1	1	1
X	X	X	X	X	X	X	1	0	X	1	0	0	0
X	X	X	X	X	X	X	X	1	X	1	0	0	1

$$A = I_8 I_9' + I_8$$

$$B = I_7 I_8' I_9' + I_6 I_7' I_8' I_9' + I_5 I_6' I_7' I_8' I_9' + I_4 I_5' I_6' I_7' I_8'$$

$$C = I_2 I_3' I_4' I_5' I_6' I_7' I_8' I_9' + I_3 I_4' I_5' I_6' I_7' I_8' I_9' + I_7 I_8' I_9' + I_6 I_7' I_8' I_9'$$

$$D = (I_1 I_2' I_3' + I_3) I_4' I_5' I_6' I_7' I_8' I_9' + (I_5 I_6' I_7' + I_7) I_8' I_9' + I_9$$